JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-10
[2018-P1]
KEY SHEET

PHYSICS

1	**AC**	2	**CD**	3	**AC**	4	**AC**	5	**ABC**	6	**ABD**
7	**0.82–0.83**	8	**6.93**	9	**1.81**	10	**1.5**	11	**0.8**	12	**1.5**
13	**800**	14	**168**	15	**A**	16	**B**	17	**D**	18	**B**

CHEMISTRY

19	**A**	20	**ACD**	21	**AC**	22	**ACD**	23	**BD**	24	**BD**
25	**6**	26	**3257.5**	27	**2**	28	**8.17**	29	**2.50**	30	**36**
31	**7**	32	**3**	33	**B**	34	**A**	35	**A**	36	**C**

MATHEMATICS

37	**ACD**	38	**BC**	39	**ABC**	40	**AB**	41	**ABC**	42	**ABC**
43	**401**	44	**7**	45	**10.07**	46	**321**	47	**3**	48	**15**
49	**2**	50	**4**	51	**B**	52	**C**	53	**C**	54	**D**

SOLUTIONS
PHYSICS

1. Minima will be formed at O, if $SS_1 + S_1O - SS_2 - S_2O = \frac{n\lambda}{2}, n = 1,2,.....$

For minimum value of $d, n = 1$ $\quad\quad$ $\therefore \frac{\lambda}{4} = \sqrt{1+d^2} - 1$

$\left(1+d^2\right)^{1/2} - 1 = \frac{\lambda}{4}$

$1 + \frac{d^2}{2} - 1 = \frac{\lambda}{4}$ (Neglecting smaller terms)

$\Rightarrow d = \sqrt{\frac{\lambda}{2}}$; $\quad\quad$ $\beta = \frac{D\lambda}{d} = \sqrt{2\lambda}$

2. $\delta_1 + \delta_2 > 0$ $\quad\quad$ $\Rightarrow \mu_1 > \mu_2$

$\theta > 0$ $\quad\quad$ $\theta_1 - \theta_2 > 0$

3. $E = \frac{R}{2}\frac{dB}{dt}$ $\quad\quad$ Impulse $= \int Fdt = \int qEdt = \int \frac{qR}{2} dB = \frac{qRB}{2}$ $mv = \frac{qRB}{2}$

Minimum velocity to complete a circle $= \sqrt{4gR}$

$q = \frac{2mv}{RB} = \frac{2m}{RB}\sqrt{4gR} = \frac{4m}{B}\sqrt{\frac{g}{R}}$

4. $\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B = 0$ at $t = t_0$ $\quad\quad$ $\Rightarrow N_A = \frac{\lambda_2}{\lambda_1} N_B = \frac{\lambda_2}{\lambda_1} \times \frac{3N_0}{2}$

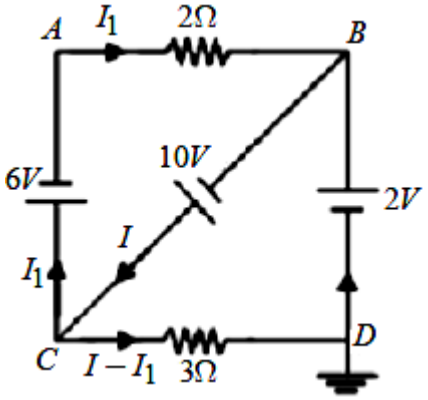
Also, $\lambda_1 (2N_0) e^{-\lambda_1 t_0} = \lambda_2 \frac{3N_0}{2}$ $\quad\quad$ $e^{-\lambda_1 t_0} = \frac{3\lambda_2}{4\lambda_1}$ $\quad\quad$ $t_0 = \frac{1}{\lambda_1} \ell n \frac{4\lambda_1}{3\lambda_2}$

5. $V_C - 6 - 2I_1 - 2 + 3(I - I_1) = V_C$

$8 = 3I - 5I_1 ------ (i)$

$V_C - 3(I - I_1) + 2 + 10 = V_C$

$3I - 3I_1 = 12 ------ (ii)$



From eqn. (i) and (ii),

$I_1 = 2A$

$I = 6A$

$I - I_1 = 4A$

Also current in wire D to earth is 0.

6. $f_{\min} = \mu mg \qquad a_{boat} = \dfrac{\mu mg}{M}$

a_{man} w.r.t boat = $\mu g + \dfrac{\mu mg}{M} = \dfrac{\mu(M+m)}{M} g$

$\ell = \dfrac{1}{2} \dfrac{\mu(M+m)g}{M} t^2$

$v = 0$ (no external force)

7. $T(x) = T + \dfrac{2T - T}{\ell} x = (1+x)\dfrac{T}{\ell}$

$v(x) = \sqrt{\dfrac{\gamma RT(x)}{M}} = \sqrt{\dfrac{\gamma RT}{M\ell}(1+x)} = \dfrac{dx}{dt}$

$t = \int_0^t dt = \sqrt{\dfrac{M\ell}{\gamma RT}} \int_0^{\ell} \dfrac{dx}{\sqrt{\ell + x}} \qquad = 2\sqrt{\dfrac{M\ell}{\gamma RT}} \left(\sqrt{2\ell} = \sqrt{\ell}\right) = 2\left(\sqrt{2} - 1\right)\sqrt{\dfrac{M\ell}{\gamma RT}}$

8. $\dfrac{10}{4} = \dfrac{10}{2}\left(1 - e^{-t/\tau}\right); \tau = \dfrac{L}{R} = 10\,s \;\; \therefore t = 10 \times \ell n\, 2$

9. Least court $(LC) = 1\,MSD - 1\,VSD = 0.05\ mm - \dfrac{9}{10}(0.05) mm = 0.005\ cm$

Reading of scale = Diameter = MSR + n (Least count) – zero error

$= \left[1.8 + 6 \times (0.005) - 4 \times (0.005)\right] cm \qquad D = 1.81\ cm$

10. $\vec{r}_{cm} = \dfrac{a}{2}\hat{i} + \dfrac{a}{2}\hat{j}$ (considering A to be origin AD to x–axis and AB to be y–axis)

$\vec{L}_i = m\vec{r}_{cm} \times \vec{v}_{cm} = m\left(\dfrac{a}{2}\hat{i} + \dfrac{a}{2}\hat{j}\right) \times v_0\left(2\hat{i} + 4\hat{j}\right) \qquad = mav_0\left(2\hat{k} - \hat{k}\right) = av_0\hat{k}$

$L_f = \left[\dfrac{m}{12}\left(a^2 + a^2\right) + m\dfrac{a^2}{2}\right]\omega = \left(\dfrac{ma^2}{6} + \dfrac{ma^2}{2}\right)\omega$

From conservation of angular momentum about point A

$L_i = L_f \qquad \dfrac{2}{3}ma^2\omega = mav_0 \qquad \Rightarrow \qquad \omega = \dfrac{3v_0}{2a}$

$|v_{cm}| = \omega\dfrac{a}{\sqrt{2}} = \dfrac{3v_0}{2a}\dfrac{a}{\sqrt{2}} = \dfrac{3v_0}{2\sqrt{2}}$

11. $y \to 0 \;\; \mu = 2 \;\; \dfrac{1}{f} = (2-1)\left(\dfrac{2}{R}\right) \qquad \dfrac{1}{f} = \dfrac{2}{R}$

$\dfrac{1}{v_1} + \dfrac{1}{R} = \dfrac{2}{R} \qquad \dfrac{1}{v_1} = \dfrac{1}{R}$

$\dfrac{1}{v_2} + \dfrac{1}{R} = \dfrac{6}{R} \qquad v_2 = \dfrac{R}{5} \qquad$ Image length = $\dfrac{4R}{5}$

12. Velocity of constant point on rod CD and ring should have same component of velocity along normal to rod CD. $\therefore \quad BC\left(\omega'\right) = 2\left(BC\cos\theta\right)\omega\cos\theta$

$\Rightarrow \quad \omega' = 2\omega\cos^2\theta = 3/2\ rad/s$

13. If the current in the solenoid is i, the magnetic field inside the solenoid is $B = \mu_0 ni$ parallel to its axis. Outside the solenoid, the field will be zero. The flux of the magnetic field through the coil will be $\phi = B\pi r^2 N$ where r is the radius of the solenoid and N is the number of turns in the coil. The induced emf will have magnitude.

$$\frac{d\phi}{dt} = N\ \pi r^2 \frac{dB}{dt} = \pi r^2 N\mu_0 n\frac{di}{dt}$$

If R denotes the resistance of the coil, the current through the galvanometer is

$$I = \frac{\pi r^2 N}{R}\mu_0 n\frac{di}{dt} \text{ or } Idt = \frac{\pi r^2 N}{R}\mu_0 n\ di.$$

The total charge passing through the galvanometer is

$$\Delta Q = \int Idt = \frac{\pi r^2 N}{R}\mu_0 n\int di = \frac{\pi r^2 N\mu_0 n}{R}\Delta i$$

$$= \frac{\pi\left(2cm\right)^2 \times 100 \times 4\pi 10^{-7}\ \frac{T-m}{A} \times 100\,cm^{-1} \times 10A}{20\Omega} = 8\times 10^{-4} C = 800\,\mu C$$

14. Use conservation of M.E and conservation of angular momentum about O.

15. $F = ML\omega^2$

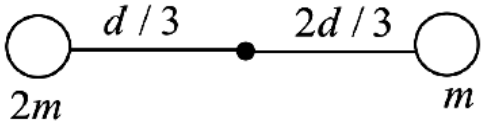
$$F = yA\frac{\Delta L}{L}$$

$$y = \frac{mL^2\omega^2}{A\Delta L}$$

16. $F = ML\omega^2$

Braking stress = Braking force/Area

17.



$$m\omega^2\frac{2d}{3} = \frac{Gm\left(2m\right)}{d^2}$$

$$T = \sqrt{\frac{4\pi^2 d^3}{3Gm}}$$

18. $$\frac{L_m}{L_{2m}} = \frac{I_m\omega}{I_{2m}\omega} = \frac{m\left(\frac{2d}{3}\right)^2}{2m\left(\frac{d}{3}\right)^2} = 2:1$$

CHEMISTRY

19. (A) Correct – Azides (such as NaN_3 and $Pb(N_3)_2$) decompose explosively to produce N_2 gas. Example:

$2NaN_3 \rightarrow 2Na + 3N_2$

$Pb(N_3)_2 \rightarrow Pb + 3N_2$

(B) Incorrect – Not all ammonium salts release N_2 gas. For example, ammonium nitrate (NH_4NO_3) decomposes to N_2O and H_2O, not directly to N_2 at lower temperatures.

(C) Incorrect – Metal nitrides react with water to release ammonia (NH_3).

(D) Incorrect – Inorganic compounds (e.g., azides, ammonium dichromate) can also produce N_2 gas upon decomposition.

20. (A) Correct: Metal carbonyl bonding is characterized by σ-donation (from CO to metal) and π-back bonding (from metal d-orbitals to CO's π^*-orbitals), leading to synergic bonding.

(B) Incorrect: An increase in metal oxidation state reduces electron density, weakening π-back bonding, which weakens the metal–carbon bond.

(C) Correct: Reduced π-back bonding in higher oxidation states strengthens the C–O bond, increasing its IR stretching frequency.

(D) Correct: The 18-electron rule is a key stability criterion for transition metal complexes, including carbonyls like $Fe(CO)_5$ and $Ni(CO)_4$.

21. **Reaction i:** Formation of **ammonium carbonate**

$$2NH_3 + H_2O + CO_2 \rightarrow (NH_4)_2CO_3 \quad (W)$$

Reaction ii: Formation of **ammonium bicarbonate**

$$(NH_4)_2CO_3 + H_2O + CO_2 \rightarrow 2NH_4HCO_3 \quad (X)$$

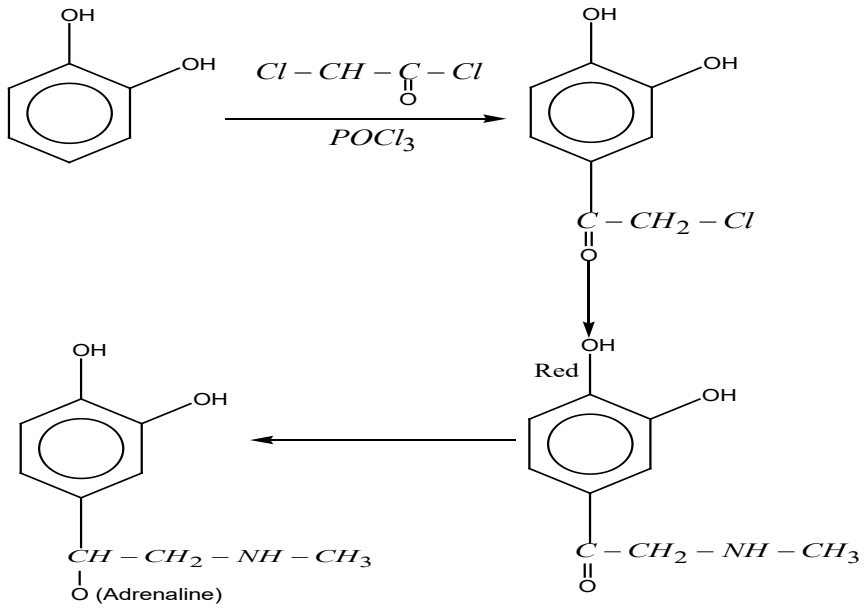
Reaction iii: Formation of **sodium bicarbonate (M) and ammonium chloride (N)**

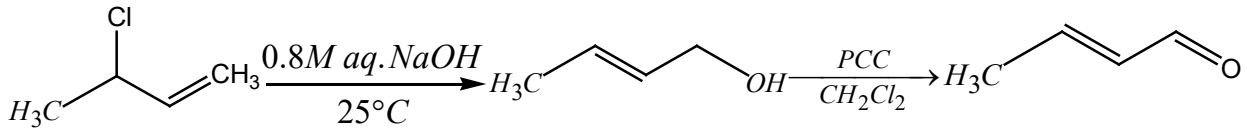
$$NH_4HCO_3 + NaCl \rightarrow NaHCO_3 \quad (M) + NH_4Cl \quad (N)$$

Reaction iv: Thermal decomposition of sodium bicarbonate

$$2NaHCO_3 \rightarrow Na_2CO_3 \quad (P) + CO_2 \quad (Q) + H_2O \quad (R)$$

22.



23. 

24. a) From Arhenius equation $K = Ae^{-Ea/RT}$

$K \propto e^{-1/T}$

Hence, K v/s T gives exponential graph

d) $\pi = CRT$

$\pi \propto T$

25. According to VBT (a), (f)f are diamagnetic

26. **Step 1: Reaction 1- Production of Ammonia**

Urea reacts with sodium hydroxide as follows:

$$\left(NH_2\right)_2 CO + 2NaOH \rightarrow 2NH_3 + Na_2CO_3$$

Molecular Weights:

1) Urea $\left(NH_2\right)_2 CO = \left(2\times14\right)+\left(4\times1\right)+12+\left(16+1\right)=60g\ /\ mol$

2) Sodium Hydroxide (NaOH) = 23 + 16 +1 = 40g/mol

3) Sodium carbonate $\left(Na_2CO_3\right)=\left(2\times23\right)+12+\left(3\times16\right)=106g\ /\ mol$

From the reaction:

1) 60g of urea produces $2\times17=34g$ of NH_3 and 106g of Na_2CO_3.

Using the unitary method for 1200g of urea:

1) Ammonia produced $=\frac{34}{60}\times1200=680g$

2) Sodium carbonate produced $=\frac{106}{60}\times1200=2120g$

Step 2: Reaction 2- Formation of Copper-Ammonia Complex

The ammonia reacts with copper(II) sulfate pentahydrate:

$$CuSO_4.5H_2O + 4NH_3 \rightarrow \left[Cu\left(NH_3\right)_4\right]SO_4 + 5H_2O$$

Molecular Weights:

1) $CuSO_4.5H_2O = 63.5+32+4\left(4\times16\right)+\left(5\times\left(2+16\right)\right)=249.5g\ /\ mol$

2) Tetraamminecopper(II) sulfate

$\left(\left[Cu\left(NH_3\right)_4\right]SO_4\right)=63.5+\left(4\times17\right)+32+\left(4\times16\right)=245.5g\ /\ mol$

From the reaction:

1) 249.5g of $CuSO_4.5H_2O$ reacts with $4\times17=68g$ of NH_3 to produce 245.5g of $\left[Cu\left(NH_3\right)_4\right]SO_4$.

Using the unitary method for 1249 g of $CuSO_4.5H_2O$:

1) Copper coordination compound produced $=\frac{245.5}{249.5}\times1249=1229g$

Step 3: Total Mass Calculation

The total mass of products formed:

Total mass = Sodium Carbonate + Copper-Ammonia Coordination Compound = 2120 +1229 = 3349g

Thus, the find answer is 3349g.

27. **Step 1: Contribution of Atoms in an FCC Unit Cell**

1) Corner atoms: Each corner atom contributes 1/8 of itself to the unit cell.

2) Face-centered atoms: Each face-centered atom contributes 1/2 of itself to the unit cell.

Total contribution:

1) Corners: $8 \times \frac{1}{8} = 1 atom$ **.**

2) Face-centers: $6 \times \frac{1}{2} = 3 atoms.$

3) Thus, the total number of atoms per unit cell = 4.

Step 2: Fraction of Atoms of A and B

Contribution of A:

1) 40% of corner atoms: $1 \times 0.4 = 0.4$
2) 60% of face-centered atoms: $3 \times 0.6 = 1.8$
3) Total A atoms = 0.4 + 1.8 = 2.2

Contribution of B:

1) 60% of corner atoms: $1 \times 0.6 = 0.6$
2) 40% of face-centered atoms: $3 \times 0.4 = 1.2$
3) Total A atoms = 0.6 + 1.2 = 1.8

Step 3: Finding Integer Values of x and y

We need to find inter values of x and y while maintaining the same ratio of A:B = 2.2:1.8.

Multiply by 5 to convert to integers:

$$A = 2.2 \times 5 = 11$$
$$B = 1.8 \times 5 = 9$$

Step 4: Compute $|x - y|$

$$|x - y| = |11 - 9| = 2$$

28. **Step 1: Calculate Total Charge Passed**

$$Q = I \times t$$

Since time is given in hours, convert it to seconds:

$$t = 0.965 \times 3600 = 3474 \text{ seconds}$$
$$Q = 80 \times 3474 = 277920 C$$

Step 2: Find Moles of Electrons Transferred

Using Faraday's law:

$$\text{Moles of electrons} = \frac{Q}{F} = \frac{277920}{96500} = 2.88 moles$$

Step 3: Find Moles of Methane (CH_4) **Required**

The oxidation reaction of methane in the fuel cell:

$$CH_4 + 8OH^- \rightarrow CO_3^{2-} + 6H_2O + 8e^-$$

From the reaction, 1 mole of CH_4 donates 8 moles of electrons.

Thus, moles of CH_4 required:

$$\frac{2.88}{8} = 0.36 moles$$

Step 4: Calculate Volume of CH_4 at STP

Using the molar volume at STP:

$$\text{Volume} = \text{moles} \times V_m$$
$$= 0.36 \times 22.7$$
$$= 8.172L$$

29. $\Delta U = 0$

$q_{rev} = -w_{rev} = 1750KJ$

$\Delta S_{sys} = \frac{1750}{300} KJ / K$

$\Delta S_{surr} = \frac{-1000}{300} KJ / K$

$\Delta S_{Total} = \frac{750}{300} = \frac{7.5}{3} = 2.50KJ / K$

30. $P_B^0 = 30 \qquad P_A^0 = 40$

$n_A = \frac{W}{80} \qquad n_B = \frac{W}{120}$

$x_A = \frac{\frac{w}{80}}{\frac{w}{80} + \frac{w}{120}}$

$= \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{3}} \qquad \frac{\frac{1}{2}}{\frac{5}{6}} \qquad x_A = \frac{3}{5}$

$P_{Total} = 40\left(\frac{3}{5}\right) + 30\left(\frac{2}{5}\right)$

$= 24 + 12 = 36$

31. For $Al(OH)_3$ to $ppt\left(OH^-\right) = 2 \times 10^{-8}$ for $Zn(OH)_2$ to $ppt\left[OH^-\right] = 10^{-6} M$ therefore $Al(OH)_3$ will ppt first

$\left[Al^{+3}\right]\left[10^{-6}\right]^3 = 16 \times 10^{-25}$

$\left[Al^{+3}\right] = 16 \times 10^{-7} M$

32. $n_{O_2} adsorbed = \frac{(1 - 0.5) \times 2.24}{0.0821 \times 273} = \frac{0.5 \times 2.24}{22.4} = 0.05$

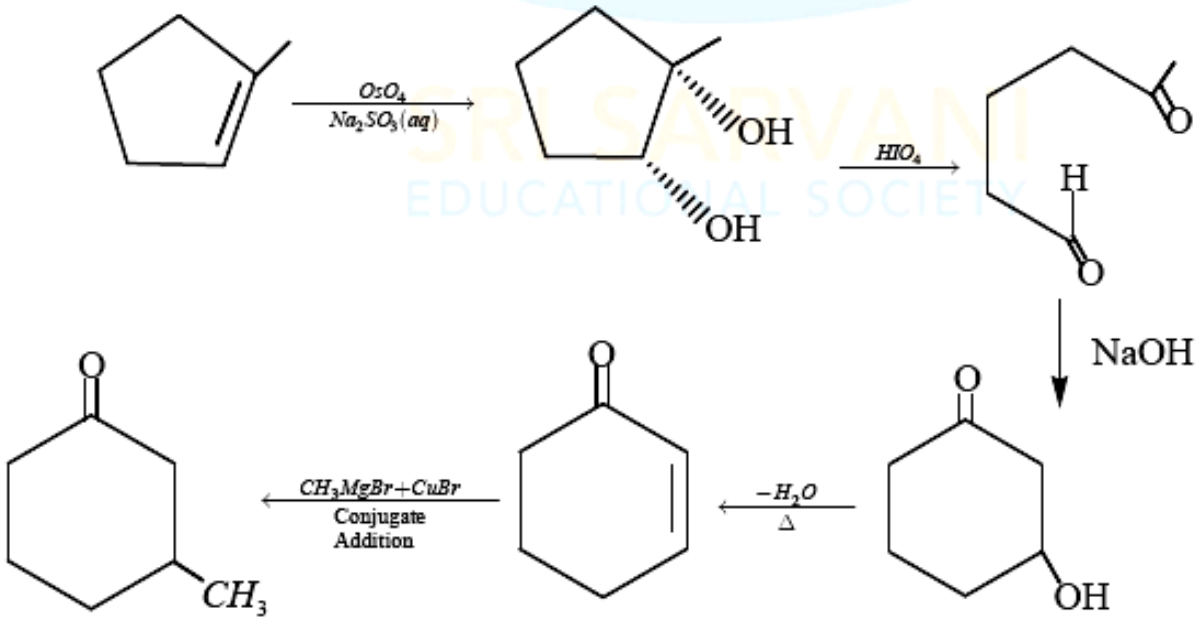
$\therefore$ no. of molecules $= 0.05 \times 6 \times 10^{23}$

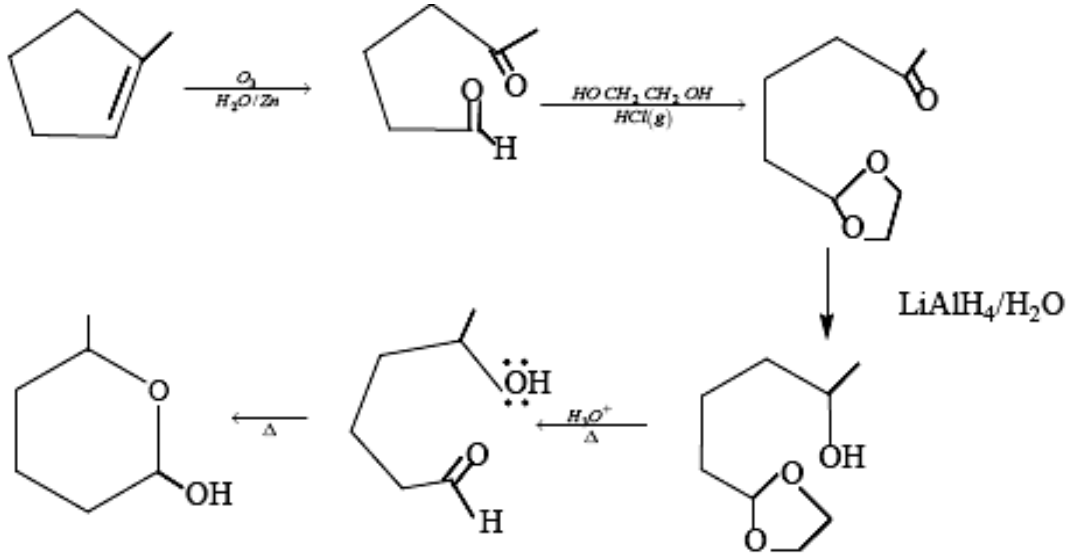
$= 3 \times 10^{22}$

$Total\ active\ sites = \left[100 \times 10^{-4}\right] \times 10^{24} = 10^{22}$

$\therefore$ no. of O_2 / *site* $= 3$

33.





34.

35&36

[C] $\rightarrow$ Benzaldehyde $\left(C_6H_5CHO\right)$

[A] $\rightarrow$ p-Aminobenzaldehyde $\left(p-NH_2C_6H_4CHO\right)$

[B] $\rightarrow$ p-Nitrobensaldiazonium salt

[D] $\rightarrow$ Cinnamaldehyde $\left(C_6H_5CH=CHCHO\right)$

[E] $\rightarrow$ p-Aminostyrene

[F] $\rightarrow$ p-Styrylbenzenediazonium salt

37. Given equation is $\vec{w}+\left(\vec{w}\times\vec{u}\right)=\vec{v}$ .....(1)

Taking cross product with $\vec{u}$, we get $\vec{u}\times\left[\vec{w}+\left(\vec{w}\times\vec{u}\right)\right]=\vec{u}\times\vec{v}$

$\vec{u}\times\vec{w}+\vec{u}\times\left(\vec{w}\times\vec{u}\right)=\vec{u}\times\vec{v}$ $\vec{u}\times\vec{w}+\left(\vec{u}.\vec{u}\right)\vec{w}-\left(\vec{u}.\vec{w}\right)\vec{u}=\vec{u}\times\vec{v}$

$\vec{u}\times\vec{w}+\vec{w}-\left(\vec{u}.\vec{w}\right)\vec{u}=\vec{u}\times\vec{v}$ .....(2)

Taking scalar product of (1) with $\vec{u}$, we get $\vec{u}.\vec{w}+\vec{u}.\left(\vec{w}\times\vec{u}\right)=\vec{u}.\vec{v}$

$\Rightarrow\vec{u}.\vec{w}=\vec{u}.\vec{v}\left[\because\vec{u}.\left(\vec{w}\times\vec{u}\right)=0\right]$ .....(3)

Taking scalar product of (1) with $\vec{v}$, we get

$\vec{v}.\vec{w}+\vec{v}.\left(\vec{w}\times\vec{u}\right)=\vec{v}.\vec{v}$ $\vec{v}.\vec{w}+\left[\vec{v}\vec{w}\vec{u}\right]=1$

$\left(\vec{u}\times\vec{v}\right).\vec{w}=1-\vec{v}.\vec{w}$ .....(4)

Taking scalar product of (2) with $\vec{w}$, we get

$\left(\vec{u}\times\vec{w}\right).\vec{w}+\vec{w}.\vec{w}-\left(\vec{u}.\vec{w}\right)\left(\vec{u}.\vec{w}\right)=\left(\vec{u}.\vec{v}\right).\vec{w}$ [From (2)]

$0+\left|\vec{w}\right|^2-\left(\vec{u}.\vec{w}\right)^2=\left(\vec{u}\times\vec{v}\right).\vec{w}$ .....(5)

or $\left(\vec{u}\times\vec{v}\right).\vec{w}=\left|\vec{w}\right|^2-\left(\vec{u}.\vec{w}\right)^2$

Taking scalar product of (1) with $\vec{w}$, we get

$\vec{w}.\vec{w}+\left(\vec{w}+\vec{u}\right).\vec{w}=\vec{v}.\vec{w}$ or $\left|\vec{w}\right|^2=\vec{v}.\vec{w}$

from eq (4), $\left|\vec{w}\right|^2=1-\left(\vec{u}\times\vec{v}\right).\vec{w}$

from eq (5), we get

$\left(\vec{u}\times\vec{v}\right).\vec{w}=\left|\vec{w}\right|^2-\left(\vec{u}.\vec{w}\right)^2=1-\left(\vec{u}\times\vec{v}\right).\vec{w}-\left(\vec{u}.\vec{w}\right)^2$

From eq (3), $2\left(\vec{u}\times\vec{v}\right).\vec{w}=1-\left(\vec{u}.\vec{v}\right)^2$

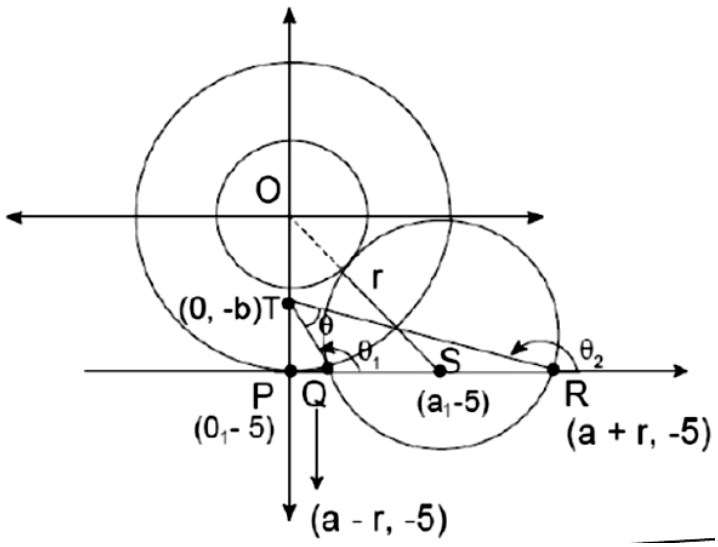
$\left|\left(\vec{u}\times\vec{v}\right).\vec{w}\right|=\frac{1}{2}\left|1-\left(\vec{u}.\vec{v}\right)^2\right|\le\frac{1}{2}\left[\because\left(\vec{u}.\vec{v}\right)\ge0\right]$

The equality holds if and only if and only if $\vec{u}.\vec{v}=0$ or iff $\vec{u}$ perpendicular to $\vec{v}$.

38. $\tan\theta=\dfrac{\dfrac{-5+b}{a+r}-\dfrac{-5+b}{a-r}}{1+\dfrac{(b+5)^2}{a^2-r^2}}$

Now, $a^2=(r+4)^2-5^2=r^2-9+8r$

$(b-5)^2=9$



39. Consider $f'''(x) = g'''(x), \forall x \in [0,3]$

On integrating, we get $f''(x) = g''(x) + C_1$

For, $x = \frac{1}{2}$, we get

$f''\left(\frac{1}{2}\right) = g''\left(\frac{1}{2}\right) + C_1 \Rightarrow 0 = 0 + C_1 \quad \Rightarrow f''(x) = g''(x) \ldots (i)$

On integrating Eq. (i) w.r.t.x, we get $\Rightarrow f'(x) = g'(x) + C_2$

For x = 1, we get

$f'(1) = g'(x) + C_2 \Rightarrow f'(1) = g'(1) + C_2$

$\Rightarrow 4 = 2 + C_2 \quad \Rightarrow C_2 = 2 \quad \Rightarrow f'(x) g'(x) + 2 \ldots\ldots\ldots (ii)$

On integrating Eq. (ii), we get $f(x) = g(x) + 2x + C_3$

For x = 3, we have

$f(3) = g(3) + 6 + C_3 \quad \Rightarrow 12 = 3 + 6 + C_3 \quad \Rightarrow C_3 = 3$

$\Rightarrow f(x) = g(x) + 2x + 3$

(a) clearly, f (2) = g (2) + 4 + 3

(b) $|f(x) - g(x)| < 5 \quad \Rightarrow |2x + 3| < 5 \quad \Rightarrow -5 < 2x + 3 < 5$

$\Rightarrow -8 < 2x < 2 \quad \Rightarrow -4 < x < 1 \quad \Rightarrow x \in (-4,1)$

(c) $[f(x) - g(x)] > 5 \quad \Rightarrow x \in (-\infty, -4) \cup (1, \infty)$

40. $y' = \frac{(y - x) \pm \sqrt{(x - y)^2 + yxy}}{2y} \qquad \frac{dy}{dx} = \frac{y - x \pm (x + y)}{2y} = 1 \text{ or } -\frac{x}{y}$

41. Let the roots of $az^3 + bz^2 + cz + d = 0, z_1 = x_1, z_2, z_3 = x_2 \pm iy_2$

$\Rightarrow z_1 + z_2 + z_3 = -\frac{b}{a} \quad \Rightarrow ab > 0$

Also $z_1 z_2 z_3 = x_1 \left(x_2^2 + y_2^2\right) = -\frac{d}{2} \quad \Rightarrow ad > 0$

42. $2ae = 5, 2a = \left(2\sqrt{2} + 1\right)\sqrt{5}, e \Rightarrow \frac{\sqrt{5}}{2\sqrt{2} + 1}$

Foci are $S_1(1,1)$ and $S_2(4,5)\dfrac{S_1N}{S_2N}=\dfrac{PS_1}{PS_2}=\dfrac{\sqrt{40}}{\sqrt{5}}=2\sqrt{2}$

43. $f(x)=f(4-x), f(x)=f(x+10), f(x)=f(14-x)$. There are 201 multiples of 10 and 200 numbers that give remainder 4 when divided by 10.

44. Put $7\theta=2n\pi, n\in z$

$\Rightarrow 4\theta=2n\pi-3\theta$

$\Rightarrow \quad \sin 4\theta=-\sin 3\theta$

$4\sin\theta\cos\theta\left(1-2\sin^2\theta\right)=-3\sin\theta+4\sin^3\theta$

$\Rightarrow 16\left(1-\sin^2\theta\right)\left(1-4\sin^2\theta+4\sin^4\theta\right)=16\sin^4\theta=24\sin^2\theta+9$

$64\sin^6\theta-112\sin^4\theta-56\sin^2\theta-7=0$ — $\sin^2 2\pi/7$, $\sin^2 4\pi/7$, $\sin^2 8\pi/7$

Hence, $\sin^2\dfrac{2\pi}{7}+\sin^2\dfrac{4\pi}{7}+\sin^2\dfrac{8\pi}{7}=\dfrac{112}{64}=\dfrac{7}{4}$

45. Let X_0 be initial population of the country and Y_0 be its initial food production. Let the average consumption be a unit. Therefore, food required initially a X_0. It is given,

$Y_0=aX_0\left(\dfrac{90}{100}\right)=0.9aX_0$ …….(i)

Let X be the population of the country in year t

Then, $\dfrac{dX}{dt}$ = rate of change of population

$=\dfrac{3}{100}X=0.03X$

$\Rightarrow \dfrac{dX}{X}=0.03dt$

$\Rightarrow \int\dfrac{dX}{X}=\int 0.03\,dt$

$\Rightarrow X=A.e^{0.03t}$ when $A=e^c$

At t = 0, $X=X_0$, thus $X_0=A$

$\therefore X=X_0e^{0.03t}$

Let Y be the food production in year t.

Then, $Y=Y_0\left(1+\dfrac{4}{100}\right)^t=0.9aX_0(1.04)^t$

$\because Y_0=0.9aX_0$ [From equation 9i]

Food consumption in the year t is $aX_0e^{0.03t}$

Again, $Y-X\geq 0$ (given)

$\Rightarrow 0.9X_0a(1.04)^t>aX_0e^{0.03t}$

$$\Rightarrow \frac{(1.04)^t}{e^{0.03t}} > \frac{1}{0.9} = \frac{10}{9}$$

Taking log on both sides, we get

$$t\left[\log(1.04) - 0.03\right] \geq \log 10 - \log 9$$

$$\Rightarrow t \geq \frac{\log 10 - \log 9}{\log(1.04) - 0.03}$$

46. $$\prod_{k=0}^{12}\left(2 - 2\omega^k + \omega^{2k}\right) = \prod_{k=0}^{12}\left(\omega^k - (1+i)\right)\left(\omega^k - (1-i)\right) = \prod_{k=0}^{12}\left((1+i) - \omega^k\right)\left((1-i) - \omega^k\right)$$

We consider the polynomial $x^{13} - 1 = 0$ whose roots are the 13th root of unity

Let $\omega^0, \omega^1, \omega^2, ...\omega^{12}$ are the roots of this equation

$$\Rightarrow x^{13} - 1 = \left(x - \omega^0\right)\left(x - \omega^1\right)\left(x - \omega^2\right)\left(x - \omega^3\right).....\left(x - \omega^{12}\right)$$

Put x = 1 + i and x = 1 – i

$$\left((1+i)^{13} - 1\right)\left((1-i)^{13} - 1\right) = (i)(-i)\prod_{k=0}^{12}\left((1+i) - \omega^k\right)\left((1-i) - \omega^k\right)$$

$$\Rightarrow \prod_{k=0}^{12}\left((1+i) - \omega^k\right)\left((1-i) - \omega^k\right) = \left(\left(\sqrt{2}e^{\frac{\pi i}{4}}\right) - 1\right)\left(\left(\sqrt{2}e^{\frac{-\pi i}{4}}\right)^{13} - 1\right)$$

The given expression $= (2)^{13} - \left(\sqrt{2}\right)^{13}\left[e^{\frac{-13i\pi}{4}} + e^{\frac{13i\pi}{4}}\right] + 1$

$$= 2^{13} - 2^{\frac{15}{2}}\left(-\frac{1}{\sqrt{2}}\right) + 1 = 2^{13} + 2^7 + 1 = 8321$$

47. By properties $PA = PB = \sqrt{10}$

$$\frac{x-1}{\frac{-1}{\sqrt{10}}} = \frac{\frac{y-5}{3}}{\frac{3}{\sqrt{10}}} = \pm\sqrt{10} \Rightarrow A \equiv (0,8) : B \equiv (2,2)$$

Slopes of asymptotes are – 7, and 1

If angle between asymptotes be θ then

$$\tan\theta = \left|\frac{-7-1}{1-7}\right| = \frac{4}{3}$$

Now, $2\tan^{-1}\frac{b}{a} = \tan^{-1}\frac{4}{3} \Rightarrow \frac{\frac{2b}{a}}{1 - \frac{b^2}{a^2}} = \frac{4}{3} \Rightarrow \frac{b}{a} = \frac{1}{2} \quad \left(\frac{b}{a} = 2\, rejected\right)$ ……..(1)

$$\frac{b^2}{a^2} = e^2 - I \Rightarrow e = \frac{\sqrt{5}}{2}$$

Also, area of $\Delta CAB = ab$

$$ab = \left|\frac{1}{2}\left[1(8-2) + 0(2-1) + 2(1-8)\right]\right| = 4 \qquad \text{……..(2)}$$

$$b^2 = 2, a = 2b = 2\sqrt{2}$$

length of latus rectum $= \frac{2b^2}{a} = \frac{4}{2\sqrt{2}} = \sqrt{2} = \ell$

48. $A\{30,60,90,120\} \Rightarrow n(A) = 4$

$B = (6n : n \in N, 1 \le n \le 23) - A \Rightarrow n(B) = 19$

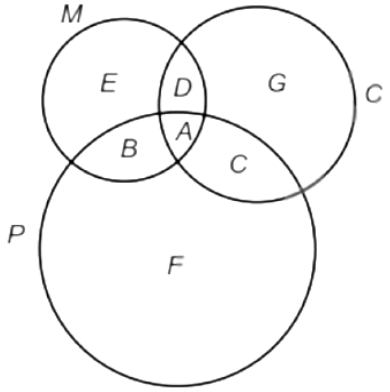
$C = (15n : n \in N, 1 \le n \le 9) - A \Rightarrow n(C) = 5$

$D = (10n : n \in, 1 \le n \le 14) - A \Rightarrow n(D) = 10$

$n(E) = 70 - n(A) - n(B) - n(D) = 70 - 33 - 37$

$n(F) = 46 - n(A) - n(B) - n(C) = 46 - 28 - 18$

$n(G) = 28 - n(A) - n(C) - n(D) = 28 - 19 - 9$



$\Rightarrow$ Number of required students = 140 – (4 + 19 + 5 + 10 + 37 + 18 + 9) = 140 – 102 = 38

49. $$\frac{1}{2013}\sum_{K=1}^{2013}\left\{1^{-} + 2K\right\} = \frac{1}{2013} \times \sum 1$$

$$\frac{e^{\tan 3n} - e^{x^3}}{2.\ell n\left(1 + x^3.\sin^2 x\right)} = \frac{e^{x^3}.\left[e^{\tan^3 n - x^3} - 1\right]}{2.\left(\tan^3 - x^3\right).\ell n\left(1 + x^3.\sin^2 x\right)}$$

$$= \frac{1}{2}.\frac{\tan^3 x - x^3}{\ell n\left(1 + x^3.\sin^2 x\right)}$$

$$=\frac{1}{2}.\frac{\left(x+\frac{x^3}{3}---\right)^3-x^3}{\frac{\ell n\left(1+x^3.\sin^{2x}\right)}{x^3.\sin^{2x}}.x^3.\sin^{2x}}$$

$$=\frac{1}{2}.\frac{\left(1+\frac{x^2}{3}---\right)^3-1}{\sin^2 x}=\frac{\frac{1}{2}\left(1+\frac{3x^2}{3}-1\right)}{\sin^2 x}=\frac{1}{2}$$

$$sum=1+\frac{1}{2}+\frac{1}{2^2}+---=2$$

50. a + b + c = 3k

$3m+3n+3p\rightarrow{}^{10}C_3$

$3m+(3n+1)+(3p+2)\Rightarrow 10.10.10$

$(3m+1)+(3n+1)+(3p+1)or(3m+2)+(3n+2)+(3p+2)\rightarrow{}^{10}C_3+{}^{10}C_3$

Total $=\cancel{3}\frac{10.9.\cancel{8}4}{\cancel{6}}+1000$

$=1360$

51. $x+y=4-z, x^2+y^2=6-z^2$

$\therefore \quad 2xy=(x+y)^2-\left(x^2+y^2\right)=(4-z)^2-\left(6-z^2\right)=2z^2-8z+10$

Quadratic equation whose roots are x and y is

$t^2-(x+y)t+xy=0$

$t^2-(4-z)t+z^2-4z+5=0$

$D\geq 0$

$(4-z)^2-4\left(z^2-4z+5\right)\geq 0$

$\Rightarrow(3z-2)(z-2)\leq 0$

$z\in\left[\frac{2}{3},2\right]$

$p=\frac{2}{3}, q=2$

$$f(x)=\frac{4x\left(x^2+1\right)}{x^2+\left(x^2+1\right)^2}=\frac{4\left(x+\frac{1}{x}\right)}{1+\left(x+\frac{1}{x}\right)^2}$$

Let $x+\frac{1}{x}=t$

$$y=\frac{4t}{1+t^2}=\frac{4}{\frac{1}{t}+t}$$

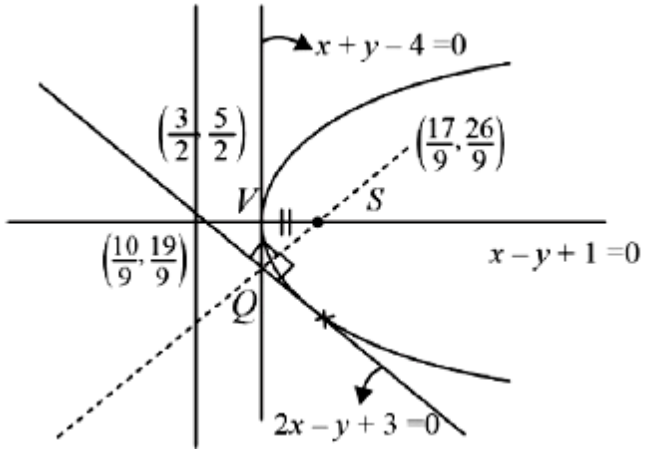
For, $x>0, t\geq 2, t+\frac{1}{t}\geq\frac{5}{2}$

$$y\in\left[0,\frac{8}{5}\right]$$

52. For exactly one root to lie in $\left(\frac{2}{3},2\right)$

$$f(2).f\left(\frac{2}{3}\right)<0 \quad \Rightarrow a\in\left(\frac{1}{3},3\right)$$

53.



Here, vertex $\equiv\left(\frac{3}{2},\frac{5}{2}\right), Q=\left(\frac{1}{3},\frac{11}{3}\right)$.

As the feet of perpendicular from focus on any tangent lies on tangent at vertex, so focus $S\equiv\left(\frac{17}{9},\frac{26}{9}\right)$.

Also, vertex is the midpoint of foot of directrix and focus.

So, foot of directrix $\equiv\left(\frac{10}{9},\frac{19}{9}\right)$

As, directrix is perpendicular to aixs, so its equation is $x+y=\lambda$

Also, $\left(\frac{10}{9},\frac{19}{9}\right)$ lies on it, so $x+y=\frac{29}{9}$ or $9x+9y=29$ (equation of directrix)

So, $l=9$ and $m=9$ Hence, $(l+m)=9+9=18$

54. We know that the length of latus rectum of parabola = 4 (Distance between vertex and focus)

$$=4\sqrt{\left(\frac{17}{9}-\frac{3}{2}\right)^2+\left(\frac{26}{9}-\frac{5}{2}\right)^2}=4\sqrt{\left(\frac{7}{18}\right)^2+\left(\frac{7}{18}\right)^2}=4\sqrt{2}\times\frac{7}{18}=\frac{14\sqrt{2}}{9}=\frac{p\sqrt{2}}{q}$$

$\therefore p=1$ and $q=9$

Hence, (p + q) = 14 + 9 = 23